

Introduction

Learning Skill Highlight: Transparency

This module's learning skill is **Transparency**. In our course, transparency turns **academic integrity** into everyday practice. Science isn't just about numbers; it's about values that guide how we do our work. Two values especially matter in this learning skill.

1. **Explanatory Power** means you can account for results. It means being able to accurately and knowingly explain relationships and mechanisms. It is not enough to be correct. It is correctly and knowingly being able to explain why you are correct. For example, if you can't read the R code that produces the statistical output because you copied what AI produces, you lose the explanatory power of the results. AI is not a researcher. AI is a black box, or **algorithmic opacity**. An AI system's internal logic is very complex, hidden, or "proprietary," making it difficult to inspect. You can't see how inputs become outputs, which limits auditability, explanation, and reproducibility (Kisselburgh & Beever, 2022). LLMs are not conscious (Shardlow & Przybyła, 2024). We must always keep humans "in the loop" of research processes that utilize AI tools. Ultimately, you, as the researcher, are *responsible* for your work.
2. **Originality** means having ingenuity and creating new knowledge and a priori truths through cognitive achievement and learning. Being original includes respecting others' intellectual property by giving credit where credit is due and acknowledging prior work. Over-reliance on LLMs can lead to creative fixation, in which researchers may become dependent on the models' suggestions, potentially stifling their independent creative thinking (Kumar et al., 2025; Cheng & Zhang, 2025). In complex tasks, this reliance can diminish the originality of research outputs, as researchers may default to ideas generated by the LLM rather than developing their own (Cheng & Zhang, 2025). There are significant concerns regarding the originality and ownership of ideas generated by LLMs. Researchers must ensure that they properly attribute any contributions made by LLMs to avoid issues of plagiarism and intellectual property infringement (e.g., Thomas, 2023).
3. **Accountability** means taking responsibility for one's research; responding to criticisms of one's research, including allegations of misconduct; self-correction (carefully scrutinizing research during peer review and after publication; retracting or correcting flawed research); and attempting to reproduce results (Resnik & Elliott, 2023).

You demonstrate explanatory power and originality through transparency practices. You make your assumptions, methods, and tools visible. That includes disclosing when and how you discovered your knowledge, such as how you use generative AI. You produce original work when you formulate your work independently. In this class, this means that you type the code yourself. If you use generative AI to help you, close the viewer and type the script yourself.

Concretely:

1. Comment your code so a collaborator could reproduce your steps;

2. name the packages and choices you made;
3. state your question or hypothesis;
4. show what you tried in what order;
5. and verify your results.

If you use AI, record it. Specifically document the tool and the vibe of your prompt(s). Just as you cite any other resource, state what knowledge you took from the AI output, and how it helped or hindered your understanding. For this course, type your own code as you are learning how to put the code together. AI tools are currently poorly designed for learning code because they encourage copy-and-paste. Remember: how you interact with learning resources can either *enhance* or *diminish* your explanatory power and originality. Real learning occurs through perseverance, productive struggle, and allowing yourself time to think.

In this module, you will consider why you are here, attending college and in this course. You will identify the values you have as a student, a future professional, and a member of society.

References

- Cheng, X., & Zhang, L. (2025). [Inspiration booster or creative fixation? The dual mechanisms of LLMs in shaping individual creativity in tasks of different complexity](#). *Humanities and Social Sciences Communications*, 12(1), 1-10.
- Kisselburgh, L., & Beever, J. (2022). The Ethics of Privacy in Research and Design: Principles, Practices, and Potential. In B. P. Knijnenburg, X. Page, P. Wiseniewski, H. R. Lipford, N. Proferes, & J. Romano (Eds.), *Modern Socio-Technical Perspectives on Privacy* (pp. 395–426). Springer. <https://doi.org/10.1007/978-3-030-82786-1>
- Kumar, H., Vincentius, J., Jordan, E., & Anderson, A. (2025, April). [Human creativity in the age of llms: Randomized experiments on divergent and convergent thinking](#). In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (pp. 1-18).
- Shardlow, M., & Przybyła, P. (2024). [Deanthropomorphising NLP: can a language model be conscious?](#). *PloS one*, 19(12), e0307521.
- Thomas, P. A. (2023). [Wikipedia and large language models: perfect pairing or perfect storm?](#). *Library Hi Tech News*, 40(10), 6-8.

Activity

Transparency with R

Purpose (2 sentences):

Today, we'll connect your personal values to how you practice Transparency in R. Transparency means making your assumptions, methods, tools, and sources visible so your work is explainable and reproducible.

1) THINK (10 minutes, quietly)

1. Go to <https://www.think2perform.com/values/> and complete the value-sorting exercise.
2. Capture your **Top Values**: Take a screenshot of your top 5 values for part of the assignment submission.
3. **Jot brief notes** (bullet points are fine):
 - **Why college?** Which value(s) brought you here?
 - **Why your major?** Name one kind of *question in your field* that needs data/statistics.
 - **Transparency link**: One way your top value(s) could show up in R reports

Privacy tip: Share only what you're comfortable sharing. If screenshots are difficult, type your top 5 values instead.

2) PAIR (10 minutes)

With a partner, briefly share one of your top values and how it can guide your practice of creating transparent, reproducible code or reporting. Share the workflow actions used when completing R assignments to make transparent documentation more efficient, accurate, etc. Discuss the challenges you experience when memoing in your R reports.

3) SHARE (10 minutes)

Volunteers share their values regarding the integrity of science and how they plan to apply them in their learning. Share a strategy for memoing and how it has helped. Listen for practical ideas you can "reuse." Hint: If you decide to reuse an idea, note its source.

What to Submit

Due: At the end of class. **Credit:** 1 Learning-Skill point (Complete/Not Complete). **Format:** Textbox submission

1. **Evidence of Values:** Upload your **screenshot** of your [think2perform](#) values results (or a typed list of your top 5 values).
2. **Short Reflection (120–180 words):** In your own words, address four prompts:
 - *Why college for you right now?* Name the value(s) driving that choice.
 - *Why this major,* and how does statistics support one question in your field?
 - *Transparency plan:* One specific action you will practice in your studies.
 - *Accountability:* How this action helps you keep **explanatory power** and **originality** in your work.

If you used AI or reused an idea from discussions to help draft the reflection, add a one-line 🧐🧐 **Resource Cited Log** at the end. For AI use, record: which tool, the *vibe* of your prompt(s) and specifically how it helped or hindered your process. Ensure the final submission is in your own voice.

Specs Grading (Complete / Not Complete)

To earn **Complete (1 point)**, your submission must include all of the following:

-  **Values evidence** (screenshot or typed top values).
-  **Reflection** that answers all four prompts with specific, personal details (no generic statements).
-  **One concrete, verifiable Transparency action** you'll use on your next R assignment.
-  🧐🧐 **Resource Cited Log** for resources.

This activity is short by design—practice the habit, not perfection. Please submit before leaving class today.

